Uncertainty-Aware Heart Murmur Detection from Phonocardiogram Recordings Using Classical Machine Learning: A Feature Engineering and Screening Pipeline for Resource-Constrained Settings

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Abstract

Cardiac auscultation—listening to heart sounds with a stethoscope—is the most widely used first-line screening tool for heart disease worldwide. However, accurate interpretation requires specialist expertise that is scarce in resource-constrained settings: general practitioners achieve only 30–50% accuracy in detecting heart murmurs, and inter-observer agreement among cardiologists is only fair ($\kappa = 0.48$). This project proposes a heart murmur detection system centred on rich multi-domain feature engineering, applied to the PhysioNet/CinC Challenge 2022 dataset—the largest publicly available phonocardiogram (PCG) database, containing 5,272 recordings from 1,452 primarily pediatric patients screened in rural Brazil. Existing classical ML studies on this dataset use isolated feature families (MFCCs alone achieving 68% accuracy, wavelet features alone achieving 76%), but no study has unified all five feature domains—MFCCs, spectral descriptors, heart cycle temporal features, fractal complexity measures, and wavelet entropy—into a single framework. We engineer 35+ features from raw PCG waveforms and evaluate them using three models: Support Vector Classification with RBF kernel (SVC-RBF) as the primary classifier given its proven superiority on this dataset, Random Forest for feature importance analysis revealing which audio characteristics are most diagnostic, and Gaussian Process Classification (GPC) as an exploratory uncertainty quantification model—enabling confidence-aware predictions. To our knowledge, no prior study has applied Gaussian Process Classification to heart sound analysis, and no study has systematically combined all five feature families on the 2022 dataset. This work addresses three identified gaps: (1) the absence of unified multi-domain feature engineering for classical ML on the PhysioNet 2022 dataset, (2) the lack of systematic feature importance analysis revealing which audio characteristics drive murmur detection, and (3) the unexplored potential of Bayesian uncertainty quantification for heart sound screening.

1 Introduction

Cardiovascular diseases (CVDs) remain the leading cause of death globally, accounting for an estimated 17.9 million fatalities annually [World Health Organization, 2021]. Early detection through cardiac screening can significantly reduce mortality, yet screening requires skilled interpretation of heart sounds—a capability that is declining even among trained physicians [Mangione and Nieman, 1997, Vukanovic-Criley et al., 2006]. Heart murmurs, which are abnormal sounds caused by turbulent blood flow through diseased or malformed heart valves, are the primary auscultatory indicator of structural heart disease. Detecting these murmurs early can prompt timely referral for echocardiography and intervention.

The challenge is that murmur detection is difficult. A landmark study by Lam et al. demonstrated that primary care physicians' cardiac auscultation accuracy ranges from only 30–50%, with recognition rates of 60–80% for systolic murmurs but as low as 20–30% for diastolic murmurs [Lam et al., 2015]. Even among cardiologists, inter-observer agreement for murmur identification reaches only $\kappa = 0.48$ [Dahl et al., 2022]. Detection of the third (S3) and fourth (S4) heart sounds—critical indicators of heart failure—shows inter-observer agreement barely above chance ($\kappa = 0.18$ and $\kappa = 0.05$ respectively) [Ishmail et al., 1987]. A multi-centre study testing medical students, trainees, physicians, and faculty demonstrated uniformly low ability to recognise systolic and diastolic murmurs [Vukanovic-Criley et al., 2006].

Digital stethoscopes now enable recording, storage, and computational analysis of heart sounds (phonocardiograms, PCGs). Machine learning applied to these recordings promises to democratise cardiac screening by providing cardiologist-level expertise in settings where cardiologists are unavailable. However, existing ML approaches are predominantly deep learning-based, requiring substantial computational resources and offering

limited interpretability [Chen et al., 2021]. Classical ML approaches with handcrafted features remain underexplored, particularly on the most recent and largest public dataset (PhysioNet 2022), and no study has provided Bayesian uncertainty quantification for heart sound classification.

The central research question is: *Can classical machine learning models, trained on rich audio features engineered from phonocardiogram recordings, detect heart murmurs with clinically useful accuracy and provide calibrated uncertainty estimates suitable for deployment in resource-constrained cardiac screening?*

2 Problem Statement

In rural and underserved communities—including remote areas of Nepal, sub-Saharan Africa, and rural Brazil—cardiac screening depends entirely on the auscultation skills of general practitioners, health assistants, or community health workers. These practitioners lack the specialised training to reliably detect heart murmurs:

(a) A general practitioner detects heart murmurs with only 30–50% accuracy [Lam et al., 2015]. Nearly half of patients with genuine heart valve disease are missed during routine examination.

(b) When a murmur *is* detected, the practitioner cannot reliably determine whether it is innocent (harmless) or pathological (indicative of disease), leading to either unnecessary referrals (costly for patients and health systems) or missed diagnoses (potentially fatal).

(c) No existing tool provides the practitioner with a *confidence score*. A system that says "murmur detected with 92% confidence—refer for echocardiography" versus "murmur uncertain at 55%—re-record from a different position" would fundamentally change clinical decision-making.

This is a machine learning problem because: (a) heart sound signals contain complex spectral, temporal, and morphological patterns that cannot be captured by simple threshold rules; (b) the relationship between audio features and murmur presence is nonlinear—murmurs vary in timing, pitch, intensity, and shape depending on the underlying pathology; and (c) the goal is *prediction* on unseen patients with quantified uncertainty, not merely testing statistical associations.

3 Literature Review and Gap Analysis

3.1 Classical ML for Heart Sound Classification

Classical machine learning has been applied to heart sound classification primarily using the PhysioNet/CinC 2016 dataset. Fuadah et al. [Fuadah et al., 2023] compared KNN, Random Forest, ANN, and SVM with MFCC features and grid search hyperparameter tuning, achieving 93.47% accuracy with Random Forest on the 2016 dataset but only 68.76% on the harder 2022 dataset. A recent study on segmented vs. non-segmented classification [Authors, 2025b] achieved 99.97% accuracy with KNN on segmented EMD features from the 2016 dataset, though such results on noisy clinical data raise overfitting concerns. SVM achieved 87% accuracy when combined with signal decomposition methods [Authors, 2025c]. Fractal complexity features with KNN achieved 97.17% on a small custom dataset [Ahlstrom et al., 2008], and wavelet-based multiscale features with SVM achieved 76.61% on the 2022 dataset [Authors, 2025d].

These studies share common limitations: most use the older 2016 dataset, employ limited feature sets (typically MFCCs alone), and provide no uncertainty quantification.

3.2 Deep Learning for Heart Sound Classification

Deep learning dominates recent heart sound research. The PhysioNet 2016 Challenge winner used an ensemble of AdaBoost and CNN classifiers with 124 time–frequency features [Potes et al., 2016]. Subsequent work achieved up to 90% accuracy with 1D CNNs [Xu et al., 2018] and 91.39% AUC with LSTM combined with MFCCs [Khan et al., 2020]. A comprehensive review identified 277 deep learning studies on heart sounds [Authors, 2024]. For the 2022 Challenge, top teams achieved approximately 80% weighted accuracy for murmur detection, predominantly using CNN and transformer architectures [Reyna et al., 2023].

Deep learning approaches, while powerful, require substantial training data and computational resources, offer limited interpretability, and—critically—are poorly calibrated, producing overconfident predictions that do not reflect true diagnostic uncertainty [Guo et al., 2017].

3.3 Uncertainty Quantification in Heart Sound Analysis

The need for uncertainty quantification in cardiac auscultation ML has been explicitly identified. Xia et al.

[Xia et al., 2022] noted that "commonly used deep learning models can be poorly calibrated, yielding overconfident predictive probabilities which cannot reflect the true diagnostic confidence." They identified uncertainty quantification as an open problem requiring solution before clinical deployment.

Only a handful of studies have attempted uncertainty quantification for heart sounds. Zhang et al. [Zhang et al., 2024] applied Monte Carlo Dropout and temperature scaling to a Branchformer model on the 2022 dataset, reducing expected calibration error by 50%. A 2025 study proposed lightweight models with Monte Carlo Dropout for edge-based murmur detection [Authors, 2025a]. Both approaches use deep learning-based uncertainty estimation.

No study has applied Gaussian Process Classification—the principled Bayesian approach to uncertainty—to heart sound analysis. GPC provides mathematically grounded uncertainty estimates from the posterior predictive distribution, unlike Monte Carlo Dropout which approximates uncertainty through repeated stochastic forward passes.

3.4 Doctor Accuracy Studies

The clinical motivation for automated cardiac auscultation is supported by extensive evidence of human limitations:

- GP murmur detection accuracy: 30–50% [Lam et al., 2015]
- Systolic murmur recognition: 60–80%; diastolic: 20–30% [Lam et al., 2015]
- Inter-observer agreement (5 cardiologists, 100 patients): $\kappa = 0.48$ [Dahl et al., 2022]
- S3 detection agreement: $\kappa = 0.18$; S4: $\kappa = 0.05$ [Ishmail et al., 1987]
- Medical students: 60.3% accuracy; interns: 53.4%; residents: 68.8%; fellows: 80.1% [Kim et al., 2021]
- Multi-centre study: uniformly low murmur recognition across all training levels [Vukanovic-Criley et al., 2006]

3.5 Identified Gaps

Table 1: Research gaps addressed by this project

Gap	Description
Gap 1	No study has unified all five feature families (MFCCs, spectral, wavelet, fractal, heart cycle) for classical ML on the PhysioNet 2022 dataset. Existing studies use isolated feature families, achieving 68–77% accuracy.
Gap 2	No systematic feature importance analysis identifies which audio characteristics drive murmur detection on the 2022 dataset. It is unknown whether fractal or heart cycle features add value beyond MFCCs and wavelet features.
Gap 3	No study has applied Gaussian Process Classification for Bayesian uncertainty quantification in heart sound analysis. All existing UQ uses deep learning approaches (Monte Carlo Dropout).

3.6 Research Hypotheses

H1: The unified multi-domain feature set (35+ features) will outperform any individual feature family alone, exceeding the current best classical ML accuracy of 76.61% on the PhysioNet 2022 dataset.

H2: SVM-RBF will achieve the highest classification accuracy due to its ability to exploit nonlinear feature interactions, while Random Forest will provide complementary feature importance insights.

H3: Feature importance analysis will reveal that fractal complexity and heart cycle temporal features contribute predictive information beyond standard MFCCs and wavelet features, justifying multi-domain feature engineering.

H4: GPC uncertainty estimates (if computationally feasible) will be higher for expert-annotated "Unknown" recordings than for "Present" or "Absent" recordings, validating uncertainty quantification for ambiguous clinical cases.

4 Methodology

4.1 Dataset: PhysioNet/CinC Challenge 2022

The George B. Moody PhysioNet Challenge 2022 [Reyna et al., 2023] provides the largest publicly available phonocardiogram dataset, specifically designed for heart murmur screening in resource-constrained settings. The data was collected from primarily pediatric

patients during cardiac screening campaigns in rural Paraíba, Brazil, using the Littmann 3200 digital stethoscope.

Table 2: PhysioNet 2022 dataset summary

Attribute	Detail
Total recordings	5,272
Patients	1,452
Murmur Present	695 recordings
Murmur Absent	3,698 recordings
Murmur Unknown	879 recordings
Auscultation locations	Aortic, Pulmonic, Mitral, Tricuspid
Sampling rate	4,000 Hz
Recording duration	5–60+ seconds
Clinical outcome	Normal / Abnormal cardiac function
Demographics	Age, sex, height, weight, pregnancy status
Download	Free, no application required

The dataset includes recordings from multiple chest locations per patient, murmur annotations (Present/Absent/Unknown) by expert auscultators, and clinical outcome labels determined by comprehensive clinical examination including echocardiography.

4.2 Data Processing

1. **Audio standardisation**: All recordings resampled to a uniform 4,000 Hz. Bandpass filter (25–400 Hz) applied to remove low-frequency body sounds and high-frequency noise while preserving the heart sound frequency range.

2. **Segmentation**: Heart sounds segmented into individual cardiac cycles (S1–systole–S2–diastole) using Shannon energy envelope and peak detection, following the approach validated in Authors [2025b]. This yields 30,000–50,000 individual heartbeat segments from the 5,272 recordings.

3. **Quality filtering**: Recordings with poor signal quality (excessive noise, clipping, or insufficient heartbeat detection) are flagged and handled separately.

4. **Label assignment**: Primary target is murmur detection (Present vs. Absent). Unknown recordings are excluded from primary analysis but used to evaluate the uncertainty model's ability to flag ambiguous cases.

4.3 Feature Engineering

From each cardiac cycle or recording segment, 35+ features are engineered across six domains. The central contribution of this project is unifying feature families that have individually shown promise in separate studies but have never been combined on the 2022 dataset:

Mel-Frequency Cepstral Coefficients (26 features): MFCCs are the standard representation for audio ML, capturing the spectral envelope shape that distinguishes normal heart sounds from murmurs. Computed using librosa with 13 coefficients, plus their first-order derivatives (Δ-MFCCs) for 26 total MFCC-related features. Used alone, MFCCs achieved 68.76% accuracy on the 2022 dataset [Fuadah et al., 2023].

Spectral features (6 features): Spectral centroid (brightness/pitch centre), spectral bandwidth (frequency spread), spectral rolloff (frequency below which 85% of energy lies), spectral flatness (tonality vs. noise-like character), zero-crossing rate (signal periodicity), and RMS energy (overall loudness).

Wavelet-based features (4 features): Wavelet entropy at multiple scales, scaling exponents from discrete wavelet transform, and wavelet energy distribution across frequency bands. These features, derived from the continuous and discrete wavelet transforms, capture the multiscale structure of heart sounds. Used alone with SVM, wavelet features achieved 76.61% accuracy on the 2022 dataset [Authors, 2025d]—the current best classical result—but have never been combined with MFCCs or fractal features.

Heart cycle temporal features (6 features): S1 amplitude, S2 amplitude, S1/S2 amplitude ratio, systolic duration, diastolic duration, systolic-to-diastolic duration ratio. These directly encode the cardiac physiology: an abnormal S1/S2 ratio or prolonged systolic interval indicates valve pathology.

Fractal and complexity features (4 features): Fractal dimension (Higuchi's method), approximate entropy, sample entropy, and Hurst exponent. These capture the complexity and self-similarity of heart sound signals. Fractal features achieved 97.17% accuracy for murmur detection on a smaller dataset [Ahlstrom et al., 2008] but have never been tested on the PhysioNet 2022 data.

Signal quality features (2 features): Signal-to-noise ratio (SNR) estimate and clipping indicator. These enable the model to assess recording quality and adjust confidence accordingly.

Total: 35+ engineered features per segment from raw WAV audio, spanning all five feature families that have individually shown promise in the literature. The hypothesis is that combining these

complementary feature domains—each capturing different aspects of heart sound morphology—will exceed the performance of any individual family alone.

4.4 Model Selection and Justification

Three models are selected. Rather than a forced comparison, each model serves a distinct analytical purpose:

Model 1: SVC with RBF Kernel (Primary Classifier). SVM with RBF kernel is selected as the primary classification engine because it holds the current best classical ML result on the PhysioNet 2022 dataset—76.61% accuracy with wavelet features alone [Authors, 2025d]. The RBF kernel handles the nonlinear relationships inherent in audio data without requiring manual interaction specification. By providing SVM with our unified 35+ feature set (combining the wavelet features that drove its previous success with MFCCs, fractal, and heart cycle features), we test whether richer features improve upon the current benchmark. Hyperparameters: $C \in \{0.1, 1, 10, 100\}$, $\gamma \in \{\text{scale}, \text{auto}, 0.01, 0.001\}$.

Model 2: Random Forest Classifier (Feature Importance). Random Forest is selected specifically for its ability to rank feature importance via Gini impurity and permutation importance. This directly answers a question no prior study has addressed: *which feature families contribute most to murmur detection?* Do wavelet features dominate? Do fractal features add value beyond MFCCs? Does heart cycle segmentation help? RF achieved 68.76% on the 2022 dataset with MFCCs alone [Fuadah et al., 2023]—if our richer features improve this substantially, it validates the unified feature engineering approach. Hyperparameters: number of trees $\in \{100, 200, 500\}$, maximum depth $\in \{10, 20, \text{None}\}$, minimum samples per leaf $\in \{1, 5, 10\}$.

Model 3: Gaussian Process Classifier (Exploratory—Uncertainty). GPC is included as an exploratory model to test whether Bayesian uncertainty quantification is feasible for heart sound screening. No prior study has applied GPC to this domain. If successful, GPC would enable confidence-aware predictions ("murmur detected with 91% confidence" vs. "uncertain at 55%—re-record"). However, GPC has computational complexity $O(n^3)$, which may be prohibitive with 5,000+ recordings. If standard GPC is intractable, sparse GP approximation with inducing points will be attempted [Rasmussen and Williams, 2006]. If GPC remains computationally infeasible, this is reported as a finding, and the project relies on SVM and RF—which are independently sufficient.

4.5 Screening Pipeline

The three models form a complementary analytical framework:

1. **Signal quality check**: Assess SNR and clipping from recording. If quality is poor, prompt re-recording before classification.
2. **Feature extraction**: Compute 35+ features from the PCG recording across all six domains.
3. **Classification**: SVC-RBF predicts murmur Present/Absent (primary accuracy).
4. **Interpretability**: Random Forest identifies which feature families drive the prediction (feature importance analysis).
5. **Confidence scoring** (if GPC is feasible): GPC provides calibrated probability with uncertainty interval.
6. **Clinical decision**:
   - High confidence abnormal (>80%): Refer for echocardiography
   - High confidence normal (>80%): No referral needed
   - Low confidence (<80%): Re-record from different position, or escalate to specialist review

4.6 Experimental Design

Cross-validation: 5-fold stratified cross-validation at the patient level. No patient's recordings appear in both training and test folds.

Evaluation metrics: Accuracy, Sensitivity (recall for murmur-present class), Specificity, F1 Score, AUC-ROC. For GPC additionally (if feasible): Expected Calibration Error (ECE), Prediction Interval Coverage Probability (PICP), and reliability diagrams.

Feature ablation: Train all models on (a) MFCCs only (26 features), (b) MFCCs + spectral (32 features), (c) MFCCs + spectral + wavelet (36 features), and (d) all features including fractal and heart cycle (35+ features). This directly tests whether each additional feature family improves classification beyond MFCCs alone, and quantifies the contribution of wavelet features (which drove the current best result) versus fractal and heart cycle features (which are novel for this dataset).

Comparison with existing results: Results are compared against published classical ML results on the 2022 dataset—specifically RF with MFCCs at 68.76% [Fuadah et al., 2023] and SVM with wavelet features at 76.61% [Authors, 2025d]—as well as the 2022 Challenge deep learning leaderboard (∼80% weighted accuracy).

4.7 Implementation

All models are implemented in Python using `scikit-learn` (SVM, Random Forest, GPC), `librosa`

(MFCC and spectral feature extraction), `pywt` (wavelet transform features), `scipy` (signal processing, fractal dimension), and `numpy/pandas` (data handling). Heart cycle segmentation uses Shannon energy envelope with adaptive peak detection. Feature scaling uses `StandardScaler` fitted on training data only.

5 Expected Results

1. **Classification performance**: SVC-RBF with the unified 35+ feature set is expected to achieve 78–84% accuracy on the PhysioNet 2022 dataset, exceeding the current best classical ML result of 76.61% (SVM with wavelet features alone) by leveraging complementary information from MFCCs, fractal, and heart cycle features. Random Forest is expected to achieve 73–80%, improving on its prior 68.76% result through richer features.

2. **Feature family contribution**: Feature ablation is expected to show that the unified feature set outperforms any individual family. Specifically: MFCCs alone ≈68%, adding wavelet ≈76%, adding fractal and heart cycle ≈80%+. This quantifies the value of each feature domain.

3. **Feature importance**: Random Forest permutation importance is expected to reveal that fractal dimension and heart cycle features (S1/S2 ratio, systolic duration) contribute significant predictive information beyond standard MFCCs, justifying the rich feature engineering approach. Wavelet entropy features are expected to rank highly, consistent with their strong standalone performance.

4. **Uncertainty quantification** (if GPC is feasible): GPC predictions are expected to be well-calibrated, and uncertainty should be significantly higher for recordings labeled "Unknown" by expert annotators compared to "Present" or "Absent" recordings, validating GPC's ability to flag ambiguous cases.

5. **Clinical relevance**: Even modest improvement over the current 76% benchmark represents meaningful clinical impact. At population scale, improving murmur detection from 76% to 82% in a screening programme covering 10,000 patients would correctly identify approximately 600 additional cases of structural heart disease.

6 Ethical, Social, and Legal Considerations

Data ethics: The PhysioNet 2022 dataset was collected under institutional review board approval with informed consent. Data is de-identified and publicly released under the PhysioNet Credentialed Health Data License. No additional ethical approval is required for this secondary analysis.

Clinical responsibility: This system is a screening aid, not a diagnostic device. A "murmur detected" output should trigger referral for echocardiography, not direct treatment. The uncertainty quantification component is specifically designed to prevent overconfident false negatives that could delay diagnosis.

Paediatric considerations: The dataset is primarily paediatric. Up to 80% of children have innocent (harmless) murmurs at some point. The system must be evaluated for its ability to distinguish innocent from pathological murmurs, though the dataset does not provide this distinction directly.

Equity and access: This work specifically targets resource-constrained settings where specialists are unavailable. The choice of classical ML (rather than deep learning) is deliberate: SVM and Random Forest can run on minimal hardware, enabling deployment in settings without GPU resources or internet connectivity.

Reproducibility: All code, feature extraction pipelines, and trained models will be made publicly available.

7 Project Timeline

Week	Task
1–2	Dataset download, audio preprocessing, heart cycle segmentation, feature engineering pipeline
3	Exploratory data analysis, feature distributions, class imbalance assessment, feature correlations
4–5	Model training (SVM-RBF + RF + GPC exploratory), hyperparameter tuning, 5-fold cross-validation
6	Model evaluation, feature ablation study, feature importance analysis, GPC uncertainty analysis (if feasible)
7–8	Report writing, figures, reliability diagrams, final submission

References

C. Ahlstrom et al. Digital auscultation analysis for heart murmur detection. *Annals of Biomedical Engineering*, 36:2070–2078, 2008. URL `https://link.springer.com/article/10.1007/s10439-008-9611-z`.

Various Authors. Deep learning in heart sound analysis: From techniques to clinical applications. *Health Data Science*, 2024. URL `https://spj.science.org/doi/10.34133/hds.0182`.

Various Authors. Optimizing uncertainty-aware deep learning for on-the-edge murmur detection in low-resource settings. *arXiv preprint*, 2025a. URL https://arxiv.org/html/2511.00966v1.

Various Authors. Segmented vs. non-segmented heart sound classification: Impact of feature extraction and machine learning models. *Applied Sciences*, 15 (20):11047, 2025b. URL https://www.mdpi.com/2076-3417/15/20/11047.

Various Authors. Abnormal heart sound recognition using SVM and LSTM models in real-time mode. *Scientific Reports*, 2025c. URL https://www.nature.com/articles/s41598-025-89647-0.

Various Authors. Multiscale analysis of heart sound signals in the wavelet domain for heart murmur detection. *Scientific Reports*, 2025d. URL https://www.nature.com/articles/s41598-025-93989-0.

W. Chen et al. Deep learning methods for heart sounds classification: A systematic review. *Entropy*, 2021. URL https://pmc.ncbi.nlm.nih.gov/articles/PMC8229456/.

Lauritz Bredrup Dahl et al. Interrater and intrarater agreement on heart murmurs. *Scandinavian Journal of Primary Health Care*, 2022. URL https://www.tandfonline.com/doi/full/10.1080/02813432.2022.2159204.

Yunendah Nur Fuadah et al. An optimal approach for heart sound classification using grid search in hyperparameter optimization of machine learning. *Bioengineering*, 10(1):61, 2023. URL https://pmc.ncbi.nlm.nih.gov/articles/PMC9854602/.

Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. *International Conference on Machine Learning*, 2017.

A. A. Ishmail et al. Interobserver agreement by auscultation in the presence of a third heart sound in patients with congestive heart failure. *Chest*, 1987. URL https://www.sciencedirect.com/science/article/abs/pii/S0146280607800028. S3 kappa=0.18, S4 kappa=0.05.

F. A. Khan et al. LSTM-based heart sound classification with MFCC features. *Computing in Cardiology*, 2020. AUC 91.39%.

Yoonjoo Kim et al. Respiratory sound classification for crackles, wheezes, and rhonchi in the clinical field using deep learning. *Scientific Reports*, 11:17186, 2021. URL https://www.nature.com/articles/s41598-021-96724-7. Medical students 60.3%, interns 53.4%, residents 68.8%, fellows 80.1%.

Matthew Lam et al. Primary care physicians' cardiac auscultation accuracy. *Journal of General Internal Medicine*, 2015. URL https://www.nature.com/articles/s41598-025-04746-2. GP accuracy 30-50%, systolic 60-80%, diastolic 20-30%.

Salvatore Mangione and Linda Z. Nieman. Cardiac auscultatory skills of internal medicine and family practice trainees: A comparison of diagnostic proficiency. *JAMA*, 278(9):717–722, 1997. URL https://pubmed.ncbi.nlm.nih.gov/9286830/.

Cristhian Potes et al. Ensemble of feature-based and deep learning-based classifiers for detection of abnormal heart sounds. *Computing in Cardiology*, 2016. Winner of PhysioNet 2016 Challenge.

Carl Edward Rasmussen and Christopher K. I. Williams. *Gaussian Processes for Machine Learning*. MIT Press, 2006. URL https://gaussianprocess.org/gpml/.

Matthew A. Reyna et al. Heart murmur detection from phonocardiogram recordings: The george b. moody physionet challenge 2022. *PLOS Digital Health*, 2(9):e0000324, 2023. URL https://journals.plos.org/digitalhealth/article?id=10.1371/journal.pdig.0000324.

Janja M. Vukanovic-Criley, Sarah Criley, Carole M. Warde, et al. Competency in cardiac examination skills in medical students, trainees, physicians, and faculty: A multicenter study. *Archives of Internal Medicine*, 166(6):610–616, 2006. URL https://pubmed.ncbi.nlm.nih.gov/16567598/.

World Health Organization. Cardiovascular diseases (CVDs) fact sheet. 2021. URL https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-(cvds).

Tong Xia, Jing Han, and Cecilia Mascolo. Exploring machine learning for audio-based respiratory condition screening: A concise review. *Experimental Biology and Medicine*, 2022. URL https://pmc.ncbi.nlm.nih.gov/articles/PMC9791302/.

Y. Xu et al. A novel 1d CNN for PCG patch classification. *Computing in Cardiology*, 2018. 90.046% accuracy on PhysioNet 2016.

Various Zhang et al. Enhanced heart murmur detection via branchformer with uncertainty estimation. *arXiv preprint*, 2024. URL https://arxiv.org/html/2405.03953.